Raspberry Pi Foundation

Lesson 1: What is AI?

Share your feedback in our user survey: rpf.io/exai-2mf.
If you are an educator, ask your students to complete a short survey: rpf.io/exai-st.

Learners will be introduced to the term 'artificial intelligence' (AI) and complete activities to describe what AI is, what AI is not, and how AI systems can be used to benefit society. They will be encouraged to analyse the language that is used to describe AI and to use appropriate and technical terms.

Objectives

Objective 1	Recognise that AI is a tool that can solve a range of complex problems
Objective 2	Outline some benefits and harms of using AI applications
Objective 3	Use appropriate technical descriptions of AI systems

Keywords

- Artificial intelligence
- Generative AI
- Computer vision
- Prediction

Lesson outline

Activity	Objective	Description	Time
Activity 1: What is AI?	1	Learners explore their preexisting knowledge of the definitions, uses, and impacts of AI.	15 mins
Activity 2: Generative AI	1, 2	Learners complete a practical activity with generative AI and discuss the benefits and impacts.	15 mins
Activity 3: Describing AI applications	3	Learners practise using system-related language instead of human behaviours to describe AI applications.	10 mins
Optional: Computer vision	1, 2	Learners examine predictions output by a computer vision system and consider the benefits and risks of these systems.	~15 mins
Exit ticket	1, 2, 3	Learners review key concepts about AI by writing a short description of AI.	5 mins

Resources

- Computers with internet access
- Writing equipment to collect learners' answers in Activity 1 (for example, sticky notes, a whiteboard, or word-processing software)

Printed/Digitally distributed
- Worksheet 2: Generative AI
- Worksheet (optional): Computer vision

Links
- www.artguru.ai/ai-text-to-image-generator
- rpf.io/computer-vision (optional activity)
- rpf.io/xai-1-v1

Subject knowledge

Activity 2 and optional activity: You will need to lead discussions about the positive and negative impacts of generative AI and computer vision on society. You do not need detailed knowledge of how these technologies work, and discussion prompts are provided on the slides and in the lesson walkthrough below.

Activity 3: You will need to help learners identify ways that AI applications are described using human characteristics and behaviours. You will need to identify:
- Human behaviours like 'understands', 'listens', and 'sees'
- The term 'AI' used as a countable noun, like in "an AI", instead of using terms such as 'AI application' or 'AI system'
- AI systems described as though they have agency, like in "the AI generates text"

Assessment opportunities

Objective	Description
1	Activity 1: You can assess learners' ability to describe that artificial intelligence is used to solve complex problems after learners have watched the video and reflected on examples, impacts, and definitions of AI. Learners will also have the opportunity to explore other examples in Activity 2 and in the optional activity.
2	Activity 2: You can assess learners' understanding of some of the benefits and harms of using AI applications during the discussions in Activity 2 and in the optional activity.
3	Activity 3: You can assess learners' ability to describe AI applications using appropriate technical language.

Lesson walkthrough

The **Success criteria** on slide 3 translate lesson objectives into actionable, student friendly targets. They clearly define what the students should be able to do by the end of the lesson.

Slide deck labels appear at the top of each slide to help you navigate the activities in the lesson.

This lesson includes an **optional activity** that will take around 15 minutes. If you do not want to include it, please skip slides 17–18. These slides have been hidden by default.

15 mins	**Activity 1: What is AI?**
Slide 2	As learners enter the classroom, ask them what they think of when they hear the term 'artificial intelligence' or 'AI'. Ask them to write down their responses to share with the class.
	Educator guidance Some classes may have a lot of experience and awareness of AI already. If your class is very unfamiliar with AI, you can use the following prompts to give them ideas: ● What applications can you think of that you would call AI? ● How is AI changing the world around us? ● How would you describe AI to a friend? The aim of this activity is not for learners to give the 'correct' answer, but for you and the learners to assess their current understanding. You can take responses verbally, but it would be best to write them down so that you and the learners can reflect on them throughout the rest of the unit.
Slide 4	Ask learners to discuss with the person next to them whether the things they wrote down are **uses of AI systems**, **effects on people**, or **descriptions of AI**, and to sort them into the correct categories.
Slide 5	Explain that the term 'AI' can be applied to a lot of things. It does not relate to a coherent set of technologies.
Slide 6	The video on this slide is a compilation of experts at Google DeepMind answering the question "What should young people know about AI?"
	Educator guidance Tell learners to listen for and write down examples of the following: ● What AI is not ● What AI is Highlight that the first people in the video are members of the public, and some of their ideas may be incorrect.
	Play the video. Then, ask for the examples the learners heard of what AI is and what AI is not.
Slide 7	Ask students for feedback to the questions posed on the previous slide. Use the animations to reveal the answers on the slide. The comments on the slide are only a few examples of what was said in the video. Students may identify different comments that were made.
Slide 8	Share the definition of 'artificial intelligence'. Explain that throughout these lessons, learners will continue to explore this definition. The definition is quite vague, as more specific definitions can get outdated very quickly. The term 'AI' can be used to refer to lots of different things, as learners have just explored.

15 mins	**Activity 2: AI applications: Generative AI** **Resource required: Worksheet 2: Generative AI**
Slide 9	Give learners an overview of some of the types of problems AI applications can help us solve. Remind them of the ideas they thought of at the beginning of the lesson to help emphasise that there are lots of different types of AI systems. Then, explain that in this activity, they will focus on generative AI that can be used to create art, and they will think about the benefits and harms of this type of AI application.
	Educator guidance If you feel that the phrases "generating content" and "recommending content" are too complex for your learners, you could change "generating content" to "producing content" and "recommending content" to "suggesting content" on the slide. We have used the word 'generate' as the example refers to generative AI, and 'recommend' as the example refers to recommendation algorithms.
Slide 10	Introduce 'generative AI' and explain that it is the term used to describe artificial intelligence applications that can be used to generate content such as text, images, sound, and video.
Slide 11	Introduce learners to the AI application they are going to use to generate art (www.artguru.ai/ai-text-to-image-generator). Explain that they will need to write a prompt to generate their artwork.
	Educator guidance A prompt is a written request made to a generative AI application. The image on the slide shows an artwork generated using the prompt "Create an image that looks like a painting of the solar system. I want it to look dramatic using vivid colours." If you would like to demonstrate generating an image using www.artguru.ai/ai-text-to-image-generator, follow the steps below: • Enter a prompt • Select your preferred style of art • Click on 'Create'
Slide 12	Tell the learners that their task is to use a generative AI application to create an artwork about something that they are interested in for a class art competition. Tell them that they then need to answer the questions in **Worksheet 2: Generative AI** (the questions are also displayed on the slide). Give them a short amount of time to create their artworks and answer the questions in the worksheet. If relevant, you may wish to tell learners that they have a limited number of attempts to generate their art, as they only have a few free credits. Show some of the artwork to the class and decide on the best image as a class.

Educator guidance

- If you feel that your students will not respond well to a competition, please feel free to adapt this to a discussion where the student presents their work and the prompt they wrote. The class could discuss whether anything about the image was surprising or did not match the prompt.
- There are other AI tools for generating images, but they may require users to have accounts and may not be age-appropriate (for example, Canva, Adobe Firefly, and Craiyon).
- To use the Artguru tool:
    - Users need to be aged 13 or over. If any of your learners are under the age of 13, it is recommended that you demonstrate the tool to the class and take suggestions from learners for the prompt to use. Use the questions below to help guide discussion.
    - Users should not enter personal information into a prompt.
- Artguru is a third party that is not affiliated with the Raspberry Pi Foundation, and the link is provided for informational purposes only.
- As with any web search, inappropriate prompts or search terms may produce inappropriate results.
- Search results can take approximately 1 minute to appear.
- Learners will have a limited number of free credits and therefore a limited number of attempts to generate their art.

Feel free to adapt this task to your class. For example:
- You might want to ask learners to share their artwork with the person next to them instead of having a class competition.
- You could base the competition around a certain theme, which you could add to slide 11.

While the learners are experimenting with the art generation website, circulate around the class and encourage learners to consider the following:
- Try generating the image again with the same prompts and the same art style. Does it generate the same artwork as last time?
- Try changing your prompts. Does it work better if more or fewer words are used?
- Who might use this technology?
- How might artists feel about this technology? Do you think they might be worried?
- Who owns the artwork you created? You or the people who made the application?
- Was there anything unusual or unexpected about the image that was generated? Why do you think that happened?

---

10 mins

Activity 3: Describing AI applications

Slide 13

Read the description of a generative AI application to the learners and ask them whether the language used to describe the tool they used in Activity 2 is correct. Use the animations to prompt them to look for **human** behaviours in the description.

Use the next animation to show the learners the highlighted words — “listens”, “understand”, and “creates” — and ask them if they would use those words to describe other tools like hammers, calculators, or pencils. Explain that these words make it sound like generative AI applications have more human capabilities than they do.

Slide 14	Ask learners to select more appropriate language from the list on the slide. This slide has animations to reveal the correct answers. Be sure to highlight how these small word changes make it clear that it is a computer system that is being described and not a human. **Educator guidance** You could use **think, pair, share** for this activity. First, ask learners to **think** about how they could change the language. Then, instruct them to **pair** up and share their ideas. Finally, ask some of the learners to **share** their ideas with the class. You could also ask learners to write their answers on individual whiteboards, then share them with the class. If you feel that the terms 'processes' and 'generates' are too complex for your learners, you could change "processes" to "uses" and "generates the media" to "produces the media (like text, images, or videos)". You would need to change the wording on slide 17 as well.
Slide 15	Explain that describing AI applications with **human characteristics** can affect how users view them. Using this sort of language incorrectly identifies AI applications as human, and ignores the humans who make and use these systems.
Slide 16	Share some guidelines for describing AI applications in a way that makes it clear that they are **computer systems** that are **used and created by humans**.
15 mins	**Optional: AI applications: Computer vision** **Resource required: Worksheet: Computer vision** **Educator guidance** This is an optional extension activity. If you do not want to include it, please skip slides 12–13.
Slide 17 **Optional**	Introduce learners to the meaning of the term 'computer vision': "Computer vision is an area of AI research that concentrates on identifying data patterns in images and videos." Draw learners' attention to the example image on the slide and highlight that there are objects in the image that have been identified. Ask the question on the slide: "Why do you think there are percentages next to each object that has been identified?" The answer is that they are confidence scores, which give us an indication of how likely a prediction is to be correct. Do not spend long discussing this, as the purpose of this activity is for learners to explore what confidence scores mean.
Slide 18 **Optional**	Ask learners to follow the instructions for this activity on their worksheet. They should visit the website rpf.io/computer-vision. Once they have opened the website, learners can select one of the sample images available to see what the AI system can identify in the image.

Raspberry Pi Foundation

Allow 4 minutes for learners to experiment.

Educator guidance

While the learners are experimenting, circulate around the class and use the following questions to prompt discussion about what they are looking at:

- Why do you think there is a confidence score? Is that important?
- Why do you think the application has higher confidence scores for some elements of images than others?
- Who might use this technology?
- How important is the confidence rating for a driverless car, for example?
- How might visually impaired people benefit from this technology?
- Can you see this technology being misused?
- Would you be happy knowing you can be personally identified by a camera when you are walking around, for example?

Next, ask learners to spend a minute discussing the questions on the slide and worksheet with the person next to them. Then, take answers from the learners.

5 mins	**Exit ticket**
Slide 19	As an exit ticket for the lesson, ask learners to write down how they would describe AI to the person next to them. Their description should be one sentence long and they should be careful not to describe AI using human characteristics and behaviours.

Educator guidance

Example answers:

- “AI is the design and study of systems that appear to be intelligent.”
- “AI is a tool used to solve difficult problems.”

You may need to correct learners’ use of language in their descriptions, for example:

- “An AI” could become “an AI system”
- “This AI” could become “The system”
- “Understands” could become “identifies”
- “Create” could become “generate”

Slide 20	Review the learning objectives and look ahead to the next lesson.